

Union of sets

For sets A and B , the **union** $A \cup B$ contains the elements that are in A , in B , or in both:

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$

Intersection of sets

The **intersection** $A \cap B$ contains the elements that are in both A and B :

$$A \cap B = \{x \mid x \in A \wedge x \in B\}.$$

Disjoint sets

Two sets are **disjoint** if their intersection is empty:

$$A \cap B = \emptyset.$$

Difference of sets

The **difference** $A - B$ contains the elements in A that are not in B . This is the complement of B relative to A :

$$A - B = \{x \mid x \in A \wedge x \notin B\}.$$

Complement of a set

Let U be the universal set and $A \subseteq U$. The **complement** of A relative to U contains the elements of U that are not in A :

$$A^c = U - A = \{x \in U \mid x \notin A\}.$$

The notations $\overline{A}$ and A^c both denote this complement.